

DEMO / SAMPLE

OSINT Exposure Assessment Report

Open Source Intelligence Security Assessment

Prepared By	Sriram M
Institution	JNN Institution of Engineering
Department	Artificial Intelligence and Data Science
Assessment Type	Passive / Authorized OSINT Demonstration
Target	Fictional organization: Example Corporation
Target Domain	example-corp.test (fictional/demo)
Report Date	13 August 2026

Important Notice: This is a DEMO/SAMPLE academic report. All findings, identities, IP addresses, emails, screenshots and exposure examples are fictional or placeholders. No real credentials or private information are included.

1. Introduction & Objective

Open Source Intelligence (OSINT) is the collection and analysis of information that is publicly available through legitimate sources. In cybersecurity, OSINT helps an organization understand what an external observer can learn about domains, infrastructure, technologies, employees and public documents.

Assessment Objective

The objective is to model an authorized OSINT assessment and identify examples of information that could increase an organization's external attack surface. The assessment focuses on public metadata rather than exploitation or unauthorized access.

Scope

Scope Item	Demo Scope
Organization	Example Corporation
Domain	example-corp.test
Subdomains	www, mail, portal, dev (mock)
Infrastructure	Reserved fictional/demo IP ranges
Public information	Mock search results and metadata
Credentials	Not collected; examples are redacted

Rules of Engagement

Only publicly available or fictional information is considered. No password guessing, authentication attempts, exploitation, malware deployment or collection of real personal credentials is performed.

Expected Deliverables

The report contains methodology, evidence placeholders, findings, risk ratings and mitigation recommendations.

2. Methodology & Tools

The assessment follows a passive reconnaissance workflow. Each tool is used only to demonstrate what an authorized analyst could observe from public sources.

Tool / Technique	Purpose	Demo Output
TheHarvester	Discover indexed domains, hosts and emails	Mock host/email list
Shodan	Review publicly indexed service banners	Mock service exposure
Maltego	Map relationships between entities	Mock OSINT graph
Google Dorks	Identify indexed pages/files	Mock indexed results
WHOIS / DNS	Review registration and DNS metadata	Mock domain records

Assessment Workflow

1) Define scope → 2) Gather domain metadata → 3) Enumerate public hosts → 4) Review public technologies → 5) Search indexed content → 6) Correlate findings → 7) Assess risk → 8) Recommend mitigations.

Sample TheHarvester Finding

Type	Mock Result
Domain	example-corp.test
Hostname	mail.example-corp.test
Hostname	portal.example-corp.test
Email	employee1@example-corp.test (fictional)
Email	employee2@example-corp.test (fictional)
SCREENSHOT PLACEHOLDER Insert screenshot of authorized/demo TheHarvester terminal output here.	

3. Domain, DNS & Public Metadata Findings

Mock DNS/WHOIS Findings

Record / Metadata	Mock Value	Security Observation
A Record	192.0.2.10	Reserved demo address
MX	mail.example-corp.test	Mail infrastructure identifiable
NS	ns1.example-corp.test	Nameserver metadata exposed
TXT	SPF policy present	Positive control; review regularly
Subdomain	dev.example-corp.test	Development naming may reveal environment information

Subdomain Exposure

Subdomain	Purpose (Mock)	Risk
www	Public website	Low
mail	Email service	Low
portal	Employee portal	Medium
dev	Development environment	Medium
SCREENSHOT PLACEHOLDER Insert screenshot of demo DNS/WHOIS results here.		

Observation

Public metadata is not automatically a vulnerability. When multiple records are correlated, however, they can provide an external observer with a clearer picture of naming conventions and infrastructure.

4. Public Exposure & Technology Findings

Mock Shodan Exposure

Mock Host	Service	Technology	Risk
192.0.2.10	HTTPS/443	nginx (mock)	Low
192.0.2.11	SSH/22	OpenSSH (mock)	Medium
192.0.2.12	HTTPS/443	Apache (mock)	Low
192.0.2.13	HTTP/80	Web server (mock)	Medium
SCREENSHOT PLACEHOLDER Insert screenshot of a fictional/authorized Shodan result or instructor-provided mock screenshot here.			

Mock Indexed Content

Search Pattern	Mock Result	Assessment
site:example-corp.test	Public web pages	Expected
site:example-corp.test filetype:pdf	Old brochure PDF	Review metadata
site:example-corp.test inurl:login	Portal login page	Expected; harden access
site:example-corp.test intitle:"index of"	No sensitive result	Positive finding

Credential Exposure: No real leaked credentials were collected. A production assessment should report credential exposure through appropriate authorized channels and redact secrets from the report.

5. Maltego Correlation & Risk Assessment

Mock Maltego Relationship Map

A conceptual relationship map demonstrates how individual public observations can be correlated:

Entity	Relationship	Entity
Example Corporation	owns	example-corp.test
example-corp.test	has subdomain	portal.example-corp.test
portal.example-corp.test	resolves to	192.0.2.12
192.0.2.12	exposes	HTTPS service
HTTPS service	identifies	Mock Apache technology

SCREENSHOT PLACEHOLDER

Insert a Maltego-style relationship graph screenshot here. Use only fictional/demo entities.

Risk Rating Summary

Finding	Likelihood	Impact	Overall
Public employee email patterns	Medium	Medium	Medium
Development subdomain naming	Medium	Medium	Medium
Public service banners	Medium	Medium	Medium
Indexed old PDF metadata	Low	Low	Low
No confirmed credential exposure	Low	High	Monitor

6. Security Mitigation Recommendations

The following recommendations are based on the mock findings and represent general defensive controls.

Priority	Recommendation	Purpose
High	Remove unnecessary internet-facing services	Reduce attack surface
High	Patch and regularly review exposed software	Reduce known-vulnerability risk
Medium	Review public subdomain naming	Limit infrastructure disclosure
Medium	Audit public documents for metadata	Reduce information leakage
Medium	Use MFA for externally accessible accounts	Reduce account compromise risk
Medium	Monitor public breach notifications	Detect possible credential exposure
Low	Conduct periodic OSINT reviews	Identify newly exposed information

Recommended Security Process

Maintain an external attack-surface inventory, review DNS and certificate records periodically, remove abandoned services, enforce secure configuration baselines and establish a process for handling accidentally exposed documents or credentials.

Evidence Handling

Screenshots should contain timestamps and assessment scope where appropriate. Sensitive values must be redacted. Do not include passwords, API keys, session tokens, personal addresses or other secrets in an academic submission.

SCREENSHOT PLACEHOLDER

Insert final evidence collage / redacted screenshots here.

7. Conclusion & Evidence Checklist

Conclusion

This demonstration shows how publicly available metadata can be combined to form a useful picture of an organization's external exposure. Individual findings may appear low risk, but correlated information can improve reconnaissance capability. Regular exposure reviews, service minimization, secure configuration, patch management and careful handling of public documents can reduce this risk.

Overall Mock Exposure Rating

Category	Rating	Reason
Domain/DNS metadata	Low	Expected public infrastructure information
Email exposure	Medium	May increase phishing risk
Technology exposure	Medium	Assists external reconnaissance
Document metadata	Low/Medium	Potential information disclosure
Credentials	Not confirmed	No real credentials collected
Overall	MEDIUM	Correlated metadata warrants monitoring

Submission Evidence Checklist

Evidence	Status
Cover page with student details	✓
Scope and methodology	✓
TheHarvester screenshot placeholder	✓
DNS/WHOIS screenshot placeholder	✓
Shodan screenshot placeholder	✓
Maltego graph placeholder	✓
Google Dork/search evidence placeholder	✓
Risk assessment table	✓
Mitigation recommendations	✓

Declaration: This document is a DEMO/SAMPLE academic report. All organization names, domains, IP addresses, identities, findings and screenshots are fictional or placeholders unless separately replaced with information obtained under explicit authorization.